PS-7 run report: CDCL with an algebraic leaf, and dynamic branching

PDP oracle speedup program, `research/pdp_oracle_speedup_20261011/ps7_cdcl/` — run of 2026-10-31, 21:14–21:46 UTC

1. What ran

`ps7_cdcl/run_ps7.sh` (stages `build test graph fidelity counts verify reg clippy summary` in one lock acquisition, then the `timing` block), `examples/pdp_ps7` (binary SHA-256 in `raw/binary_hash.txt`), on the tree of `origin/main f6d0610d1` plus the PS-7 files. Host: Intel Xeon 2.10 GHz KVM guest, 4 vCPU, kernel 6.18.44-fc-v185, rustc 1.97.0; counts do not depend on the host, wall is a note. Protocol: `protocols/PS-7.md` (frozen, unedited); decisions fixed before the run: `ps7_cdcl/DESIGN.md`. Frozen inputs: the first 16 uniform targets of each PS-0 corpus (`ℓ = 8, 9, 10`; 13 refuted and 3 found per rung) and the 60 regression inputs. Arms: `a1` PS-2′s plain loop; `a1u` the same loop over the shell's one-pair leaf (control); `a2` the library solver with the leaf as lazy-clause theory; `a2p` its bare-guard encoding; `a2l` the same solver code with a lexicographic decider; `a3` the same with the re-evaluated minimum vertex cover decider. Requested and completed: 6 arms × 48 uniform targets (first model) and again with every model (full root set), 6 arms × 60 regression inputs twice, 5 arms on the 46 applicable ones, the library-versus-copy fidelity check on 48 targets, 54 timed runs; all completed, none failed. Earlier attempts (an interrupted run on another tree, a run whose every-model records had a bookkeeping defect, two failed builds) are kept under `raw/` and not used.

2. Observations

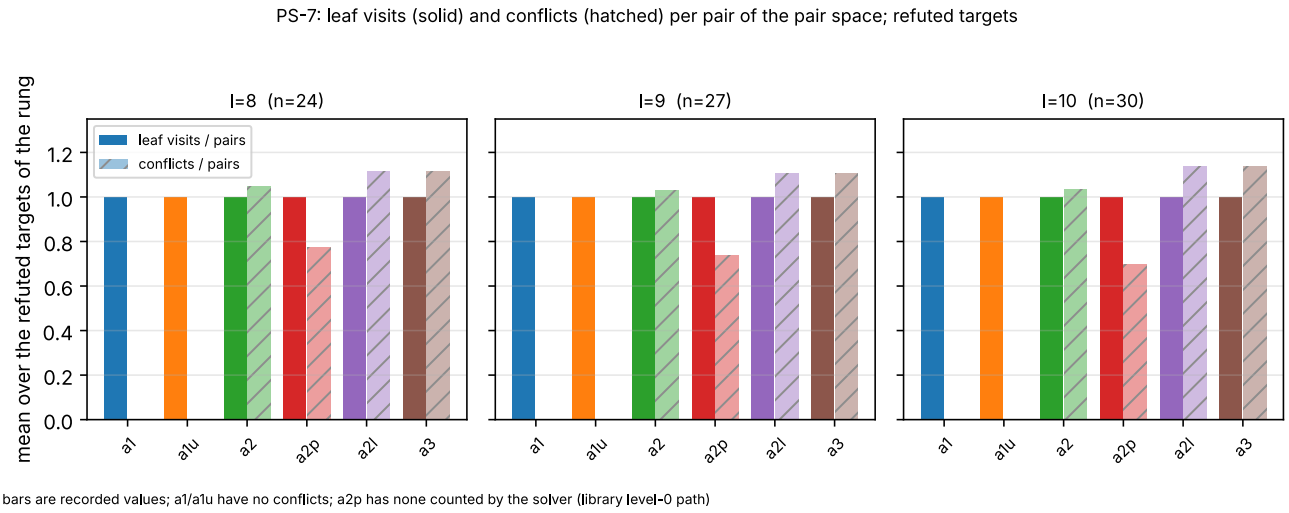


Figure 1: Leaf visits (solid) and conflicts (hatched) divided by the pair count, mean over the 13 refuted targets of each rung (13 × 3 rungs × 6 arms recorded values; source `ps7_cdcl/summary_rows.jsonl`, copy `summary_rows.jsonl` here). Leaf visits equal the pair count in every arm. `a1`, `a1u` have no conflicts; `a2p`'s solver counts none for the guards it installs without analysis.

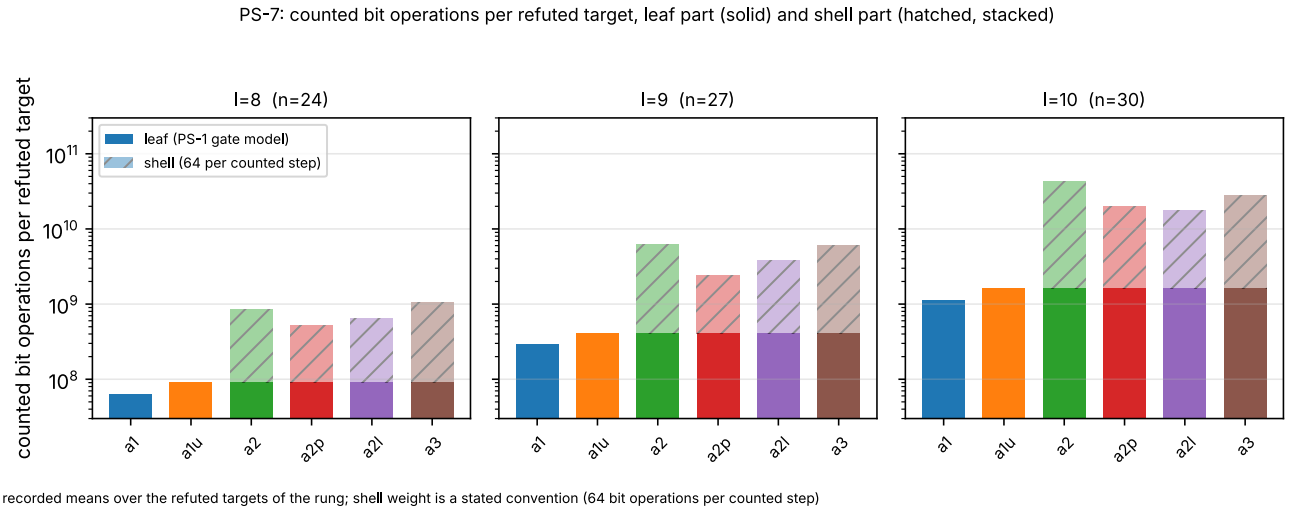


Figure 2: Counted bit operations per refuted target: leaf in PS-1′s gate model (solid) and shell at 64 bit operations per counted solver step (hatched, a stated convention). Log scale. `a1`′s leaf figure is low by about 0.5–1.6% (section 9 of the README).

ℓ	arm	pairs P	leaves L	conflicts	P / learned	total bit ops (ratio to a1)
8	a1	8,256	8,256	-	-	62,477,440 (1)
8	a1u	8,256	8,256	-	-	90,689,728 (1.45)
8	a2	8,256	8,256	8,660	0.953	852,449,984 (13.64)
8	a2p	8,256	8,256	6,369	1.000	519,230,272 (8.31)
8	a2l	8,256	8,256	9,193	0.898	635,153,792 (10.17)
8	a3	8,256	8,256	9,193	0.898	1,061,066,240 (16.98)
9	a1	32,131	32,131	-	-	294,779,914 (1)
9	a1u	32,131	32,131	-	-	406,799,206 (1.38)
9	a2	32,131	32,131	33,083	0.971	6,257,520,358 (21.23)
9	a2p	32,131	32,131	23,626	1.000	2,389,939,622 (8.11)
9	a2l	32,131	32,131	35,536	0.904	3,750,586,150 (12.72)
9	a3	32,131	32,131	35,536	0.904	6,021,078,694 (20.43)
10	a1	107,880	107,880	-	-	1,111,892,016 (1)
10	a1u	107,880	107,880	-	-	1,608,046,520 (1.45)
10	a2	107,880	107,880	111,483	0.968	42,425,100,920 (38.16)
10	a2p	107,880	107,880	75,246	1.000	19,898,423,736 (17.90)
10	a2l	107,880	107,880	122,845	0.878	17,837,296,248 (16.04)
10	a3	107,880	107,880	122,845	0.878	28,275,573,240 (25.43)

Means per refuted target (13 per rung), from `counts_summary.md`; ratios are arithmetic on the column. For `a2p` the learned clauses are the guards installed (equal to P), its conflicts are the solver's own.

Pairs removed without a leaf call: 0 on every refuted target of every rung for `a2`, `a2p`, `a2l`, `a3` (the logged per-clause maximum of such pairs is 0 over 78 records). **Pairs in the cube of a learned clause** (`a2l`, `a3`, mean): 181.7, 360.3, 610.5 at `ℓ = 8, 9, 10`: the clauses are mostly shorter than the 2ℓ pair literals (resolvents), and every pair in their cubes had already been evaluated. **Dynamic order:** the monomial graph of the descended symmetrised S_4 has only cross-summand edges ($K_{\ell,\ell,\ell}$; 75, 108, 108, 192, 243, 300 edges at $(n, \ell) = (15, 5), (17, 6), (19, 6), (24, 8), (27, 9), (30, 10)$, none inside a summand, minimum vertex cover 2ℓ); over the 48 uniform targets `a3` took 2,361,391 decisions, 0 of which differ from the lexicographic first unassigned bit, and its conflict, decision and leaf counts equal `a2l`'s. **Fidelity:** library solver against the example-local copy with hooks inactive, every counter equal on 48 of 48 targets. **Verdicts:** every record of every arm equals the PS-0 corpus' verdict (witness in the root set; every-model runs equal the full root set); regression corpus 60 of 60 (31 found including the algebraic-only input, 29 refuted), group-law leaf 46 of 46 where applicable.

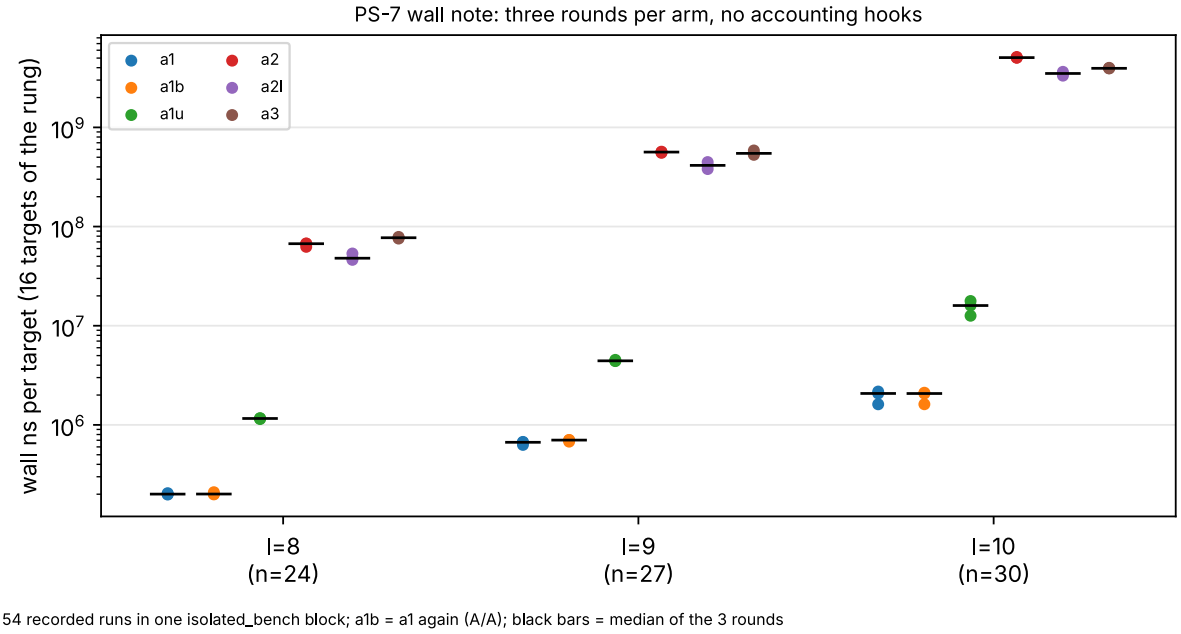


Figure 3: Wall nanoseconds per target (16 targets, found and refuted), three rounds, `a1b` = `a1` again (A/A); log scale; source `ps7_cdcl/raw/times.jsonl`. All runs inside one `isolated_bench run --cpus 3` block, not marked contended (33.3 other-CPU-seconds in 685 s); a note, not the metric.

Round-1 wall ratios to the plain loop: `a2`, `a2l`, `a3` = 315, 270, 389 at ℓ = 8; 852, 620, 819 at ℓ = 9; 2,430, 1,603, 1,900 at ℓ = 10. A/A (`a1`, `a1b`) differs by up to 12% between a pair of runs (ℓ = 9, round 3).

3. Criteria, as measured

Success (pre-registered reading: pairs removed without a leaf call, more than one per learned clause, at two rungs): not met, 0 at every rung. **Stop** (no pruning at three rungs): met, leaves = pairs on all 39 refuted targets for every realisation of arms 2 and 3. The cube reading (181.7–610.5 pairs per learned clause) exceeds 1 and corresponds to zero avoided leaf calls; both are printed in the README (section 5) and the frozen sentence does not say which is meant.

4. Missing or not done

The instrumented field-operation cross-check was done on another tree only (`raw/interrupted_ps2tree/count_check.jsonl`; the one-pair leaf's counts are exact, the plain loop's products are 0.5–1.8% low) and cannot be rebuilt on `main`, whose `semaev_decomp.rs` lacks the `opcount` module. Bit operations were not computed for the regression corpus (no gate model at $n = 15, 17, 19$). The upstream WDSat input files, WDSat/Xorricane/Bosphorus arms (out of scope in the protocol), rungs above $\ell = 10$ and theories that fire earlier than $m - 1$ summands were not run. The shell's weight (64 bit operations per counted step) is a convention; the ordering `a1` < `a1u` < every shell arm does not depend on it. The solver copy used by `a2l`, `a3` is generated from the library file (`gen_sat_hooked.sh`). Wall is one isolated block, not per-run records, on a different host class from PS-0..PS-3. No `benchmarks/` entry was added (counted-unit null result, no validation artifacts exist; see README section 10). Nothing end to end; nothing for ECC2K-130.